MARKET REQUIREMENTS DOCUMENT (MRD)

RAGNext: Autonomous Version-Aware Document Intelligence Engine with Verifiable Grounding

Problem Statement ID	SH-SVA-04 (Smart Virtual Assistant for Version-Aware Search)	**Team Name**	Cyber Knights (Sainath Loganathan, Tahir Azeem)
Target Market	Regulated Enterprises (BFSI, Healthcare, Defense, SMBs)	**Core Architecture**	Temporal Lineage Graph + Pre-Retrieval Filtering + Air-Gapped Local LLM

1. Executive Summary & Problem Framing (The Core Market Pain)

Modern enterprises operate on hundreds of evolving standard operating procedures, compliance manuals, and security guidelines. Conventional vector search engines (RAG) treat documents as flat text chunks and retrieve information based purely on semantic cosine distance. When an organization updates a policy (e.g., **Leave Policy: v1.0 [15 days] ➔ v2.0 [18 days] ➔ v3.0 [20 days]**), outdated clauses frequently score higher similarity than active ones due to exact keyword density. Standard AI systems blindly hallucinate answers from superseded documents. In heavily regulated industries (banking, healthcare, legal, defense), acting on obsolete policies leads to regulatory fines, severe audit failures, and operational disputes.

The RAGNext Solution: RAGNext is an autonomous, version-aware enterprise intelligence engine that structures documents into **Temporal Revision Lineage Graphs**, enforces **Pre-Retrieval Policy State Filtering** (ACTIVE, SUPERSEDED, DRAFT), autonomously detects **Conflicting Active Policies**, and guarantees **100% Offline Air-Gapped Data Privacy** with verifiable page citations.

2. Target Market & User Segments

Market Segment	Key Stakeholder	Core Need & Pain Addressed	Market Fit / Compliance Standard
Regulated BFSI & Defense	Compliance Officers / CISOs	Cannot risk citing deprecated policies; zero data leakage to cloud APIs.	ISO 27001 (A.5.37), NIST SP 800-53 (AU-10), CISA Zero Trust
Healthcare & Pharma	Clinical SOP Auditors	Must ensure clinical protocols in use reflect latest active drug/safety revisions.	HIPAA Privacy Rule, GxP Document Lineage
Enterprise HR & Operations	Employees / HR Leads	Resolves benefit/leave disputes; guarantees employees always see active rules.	Internal Corporate Governance & Employee Trust
Legal & Internal Audit	Regulatory Auditors	Requires bi-directional search: query historical rules (in v1.0) with verifiable citations.	Statutory Audit Defense & Dispute Reconciliation

3. Core Value Proposition & Business Impact

Metric / Value Pillar	Conventional RAG Baseline	RAGNext Expected Impact
Outdated Policy Leakage	35% - 48% citation of superseded chunks	**0% Leakage:** Hard pre-retrieval filtering eliminates inactive chunks.
Audit Turnaround Time	4 - 8 hours manual PDF diffing per audit inquiry	**60% Reduction:** Instant temporal provenance and clause citations.
Cloud Data Sovereignty	High risk: Uploads confidential IP to third-party clouds	**100% Air-Gapped:** Local Ollama LLM + embedded local vector store.
Factual Groundedness	Unverified generative text with frequent hallucinations	**> 95% Grounded:** Verbatim excerpts with exact document & page numbers.

4. User Personas (Representative Profiles)

Persona Profile	Operational Goals & Responsibilities	Current Pain Point in Unversioned Search	RAGNext Solution Feature
Rajesh Compliance & Risk Lead	Ensure policy compliance, audit historical incident records, eliminate legal exposure.	Search engines mix old and new PDF clauses; manual revision cross-checks take days.	Historical Query Intent, Conflict Alert Sentinel, and downloadable audit logs.
Priya Knowledge Worker / Employee	Get quick, authoritative answers to day-to-day policy questions (leave, VPN, benefits).	Searches return deprecated 2021 guidelines; risks disciplinary/policy violations.	Default Active Pre-Filtering with visible `[Active v3.0, Page 1]` citation badges.
Dev IT & Security Administrator	Deploy secure, private document search; maintain 100% uptime without high API costs.	Cloud AI APIs leak proprietary policies and incur unpredictable token billing.	100% offline air-gapped architecture with local Ollama and zero cloud fees.

5. User Journey Comparison: Conventional RAG vs. RAGNext

Workflow Stage	Conventional RAG Journey (Systemic Flaw)	RAGNext Version-Aware Journey (Engineered Fix)
1. Ingestion & Storage	Dumps all PDF versions into a flat vector index. Version dates and parent linkages are discarded.	Extracts SemVer, dates, and builds a directed **Revision Lineage Tree** (v1.0 ➔ v2.0 ➔ v3.0).
2. Chunking & Indexing	Splits text into arbitrary token chunks without preserving page coordinates or status flags.	**Page-Aware Chunker** retains exact page numbers, character offsets, and lifecycle status.
3. Policy Query Execution	Performs blind cosine similarity. Old v1.0 clause matches keywords and outranks active v3.0.	**Pre-Retrieval Filter** isolates search strictly to `status = 'ACTIVE'` chunks. v1.0 is excluded.
4. Historical Inquiries	Cannot filter by past dates; mixes clauses from different years into a fabricated answer.	Detects historical intent (e.g. 'in v1.0') and retrieves requested version with warning banner.
5. Conflicting Active Rules	Silently blends conflicting policies together into a nonsensical synthesis.	**Conflict Sentinel** flags an active contradiction alert and halts automated guessing.
6. Answer Grounding	Black-box AI response without traceable evidence or page verification.	Structured output with verbatim quotes, confidence ratings, and exact page citations.

6. Proposed Version-to-Action Policy Matrix

Query Intent / Trigger	Target Retrieval Scope	System Action & Output Behavior
General Policy Query	Active Version (Latest)	Hard filter `status = 'ACTIVE'`. Ignores superseded chunks. Cites active version with page proof.
Historical Query ('in v1.0')	Target Historical Revision	Retrieves target historical version; attaches prominent `■■ [Historical Record — Superseded]` warning.
Conflicting Active Policies	Multiple Active Documents	Suspends single answer. Dispatches administrative `Conflict Warning` and displays side-by-side diff.
Out-of-Scope / Missing Fact	Zero Relevant Chunks	Zero-hallucination guardrail: Returns explicit `Insufficient Evidence in Active Documents` report.

7. Requirements Matrix (MoSCoW Priorities & NFRs)

ID	Pri.	Functional Requirement Description	Target Acceptance Metric
FR01-03	**MUST**	Multi-format ingestion (PDF, DOCX, TXT), SemVer parsing, and page-aware chunking.	Preserves exact page numbers and SHA-256 hashes.
FR04-06	**MUST**	Lineage tree construction, vector indexing, and hard pre-retrieval active filtering.	100% exclusion of superseded chunks on general queries.
FR07-09	**MUST**	Historical query intent routing, conflict sentinel alerts, and grounded page citations.	Verbatim quote + page citation on every response.
FR10-12	**MUST**	Honest missing evidence reporting, web dashboard with drawer, and 1-click seed dataset.	0% hallucinations on out-of-scope questions.
NFR01-04	**NFR**	Air-gapped execution, latency <2.5s extractive / <5s LLM, dual-mode DB/AI fallback.	Zero outbound network telemetry; zero crash fallback.

8. Architectural Stack & Competitor Comparison

Solution Architecture	Version Lineage	Pre-Filter	Conflict Sentinel	Data Sovereignty	Verifiable Citations
Standard RAG (LangChain/Pinecone)	None (flat chunks)	No (cosine only)	None (silent blend)	Public Cloud API	Vague / Untraced
Enterprise Search (Elasticsearch)	None (keyword BM25)	Date filters only	None	On-Premises	Snippets only
M365 Copilot / Cloud LLMs	Basic file timestamps	Weak prompt filter	None	Public Cloud API	Document title only
RAGNext (Cyber Knights)	**Directed Tree Graph**	**Hard Pre-Filter**	**Autonomous Alert**	**100% Air-Gapped**	**Page + Quote Proof**

9. MVP Live Hackathon Demonstration Workflow & Key KPIs

Demo Step	Action Performed in UI	Expected System Output & Evaluation Criteria
1. Seed Dataset	Click 'Seed Demo Dataset' on Dashboard.	Loads 3-version Leave Policy (v1.0=15d, v2.0=18d, v3.0=20d) and Remote Work Security Policy.
2. Lineage Inspection	Open Document Library ➔ Click 'Versions'.	Displays interactive version tree showing v3.0 [Active] and v1.0/v2.0 [Superseded].
3. Current Query	Ask: 'What is the current annual leave allowance?'	Answers **20 days** citing **v3.0, Page 1**. Outdated 15/18 day clauses strictly suppressed.
4. Historical Query	Ask: 'What was the allowance in version 2?'	Answers **18 days** citing **v2.0 (Historical)** with prominent warning banner.
5. Missing Evidence	Ask: 'Where is maternity leave explained?'	Honest guardrail: Reports insufficient evidence rather than hallucinating rules.
6. Conflict Sentinel	Simulate two active documents in same group.	Dashboard triggers live Conflict Alert warning user before contradictory answers are served.

10. Business Model, Sustainability & Compliance Alignment

Monetization & Sustainability: (1) *Enterprise On-Premises License:* Annual per-node fee for air-gapped banking, healthcare, and defense deployments. (2) *Managed Private Cloud SaaS:* Single-tenant hosted instances for mid-market legal and tech firms. (3) *Turnkey Compliance Add-on:* Automated policy revision diffing and regulatory audit export reports. (4) *Open-Core Ecosystem:* Free developer core with commercial enterprise connectors (SharePoint, Google Drive).
Regulatory Compliance Alignment: Aligned with **NIST SP 800-53 Rev. 5** (Controls SI-12 Information Handling & AU-10 Audit Provenance), **ISO/IEC 27001:2022** (Control A.5.37 Documented Information Governance), and **CISA Zero Trust Data Pillar** requirements for continuous data lineage, policy integrity, and air-gapped protection.